

Classification of Glaucoma Stages Using Image Empirical Mode Decomposition and an Ensemble Learning Framework from Retinal Fundus Images

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Abstract—Glaucoma is one of the reasons people go blind all over the world. So, it is very important to find out what stage the disease is at so we can stop people's vision from getting worse. Usually doctors check for glaucoma by doing tests that need a lot of work and can be different from one doctor to another. This is why we need tools that can diagnose glaucoma automatically and always give the results. This paper talks about a way to automatically tell if a picture of the back of the eye is healthy or has glaucoma. We used an available set of pictures called RIM-ONE r12 to test our method. Our method uses a technique called Image Empirical Mode Decomposition (IEMD). IEMD is a way to break down pictures into parts that can help us understand them better. We then use these parts to get information from the pictures. We use ways to get this information, including Gray Level Co-occurrence Matrix, Local Binary Patterns, Histogram of Oriented Gradients and Gabor filter responses. We also use something called entropy measures, like Shannon, Renyi, Kapur and Yager. We get this information from the IEMD parts, the full picture and a special part of the picture called the disc. To make sure our method works well we use a way to increase the number of pictures we have. This helps us make sure our method is not closely tied to the pictures we are using. We then choose the important information and use it to teach a group of classifiers. These classifiers are like computers that can learn to tell if a picture is healthy or has glaucoma. Our method can correctly classify pictures as healthy or having glaucoma 89.43 percent of the time. It can also find pictures with glaucoma 79.51 percent of the time, and correctly identify healthy pictures 94.85 percent of the time. This is very good compared to methods that have been used before.

Index Terms—glaucoma detection, image empirical mode decomposition, ensemble learning, retinal fundus images, RIM-ONE r12, feature extraction, support vector machine

I. INTRODUCTION

Glaucoma is a progressive, lifelong disease of the optic nerve head. The disease usually is associated with increased pressure in the eye (IOP) [1]. It is estimated that 80 million people worldwide suffer from the disease, making it the second leading cause of vision loss in the world. Disorders like cataracts can be treated successfully; when glaucoma's nerve damage starts, it is permanent [2], [3]. This is very important to note as it highlights the need for early and accurate identification. There are two main types of glaucoma that eye care specialists see. Primary open-angle glaucoma (POAG) accounts for approximately 90% of all known cases. This type is slow and there are few early indications. In primary angle closure glaucoma (PACG), the fluid drainage channels become blocked suddenly, causing the symptoms to develop more quickly [4]–[7]. In POAG, the resistance within the trabecular meshwork gradually builds up, causing a progressive build-up of pressure in the eye and causing damage to the retinal nerve fiber layer over time. The conventional clinical tests use tonometers to measure IOP, ophthalmoscopes to inspect the ONH and perimeters to map visual fields [8], [9]. While these instruments are effective on their own, they are also time-consuming, require specific skills and depend on the operator's expertise to completely assess the evaluation process, which can be hard to find in resource poor areas. Retinal fundus photography is a promising option [10], [11]. This method is not invasive, is cheap and requires no expensive computer analysis, and gives a full view of the posterior part of the eye.

Many previous studies have investigated an automated method for glaucoma detection from fundus analysis [12]–[16]. On the other hand, Nayak et al. implemented the Artificial Neural Network System based on features extracted from fundus images [12]. Acharya et al. combined the Gabor transform descriptors with a least-squares support vector machine (LS-SVM) [13]. Maheshwari et al. tested the accuracy of the methods such as empirical wavelet transform (EWT) and variational mode decomposition (VMD) on the RIM-ONE r12 benchmark and achieved accuracy of 80.66% and 81.22% respectively. Later, Parashar and Agrawal have shown that their flexible analytic wavelet transform (FAWT) features gave 90.76% accuracy on the same data [17]. These results were enhanced by using enhanced extreme mode decomposition (IEMD) along with LS-SVM in their subsequent work, where they achieved 94.45% with an extra private set of data [18]. Although these developments have been made, the use of ensemble learning methods along with IEMD has not received much attention [19]. IEMD is extending the basic empirical mode decomposition to two-dimensional space by using the Hilbert-Huang transform, and decomposing images into intrinsic mode functions (IMFs) that adaptively characterize the local frequency characteristics without the need for predetermined basis functions. In this research, four contributions are made to advance the understanding in this field:

- 1) A dataset of the RIM-ONE r12 augmented by fourfold geometry.
- 2) Multi-source texture, gradient, frequency and spatial features extraction combined.
- 3) ANOVA Ranking followed by PCA compression..
- 4) An ensemble model based on soft voting of SVM, Random Forest and Gradient Boosting classifiers, with a 89.43% accuracy and 94.85% specificity on the benchmark.

II. BACKGROUND AND RELATED WORK

Researchers have been trying to create automated systems for detecting glaucoma from fundus photos for more than two decades. The first ones made use of frequency domain methods. The fast Fourier transform (FFT) features were used by Nyul and colleagues [20] and the discrete wavelet transform (DWT) energy features and SVM classification were used by Dua and others [21]. The Fourier transform, however, has rather global properties, which limit its use to detect spatial variability in structural details in fundus images. This limitation has been overcome by using wavelet-based methods, which divide the frequency plane in several divisions at different resolutions, and have been used in the development of several computer-aided diagnosis systems based on wavelets. Further improvement is achieved by using the variational mode decomposition (VMD), which decomposes the signals into band-limited intrinsic modes. The accuracy obtained by Maheshwari and colleagues is 81.22 % when using iterative VMD features with LS-SVM on the RIM-ONE r12 database [15].

The deep convolutional neural networks (CNNs) have shown to achieve remarkable performance on large scale datasets with richly annotated data. Ahn et al. created a CNN that could detect early and late-stage glaucoma with the images provided by Kim’s Eye Hospital [22]. Li and other people have developed the large-scale LAG database and an attention-enhanced detection model [23]. While deep learning techniques can yield high accuracy results, deployment in clinical environments where resources, computation resources and labeled data are scarce, continue to be challenges. This is an ideal scenario to further study handcrafted feature approaches on smaller benchmark datasets such as RIM-ONE r12.

Even with smaller number of labelled samples, traditional machine learning methods can be useful. To classify glaucoma into three classes, the higher order cumulant features were extracted from radon transforms by Noronha et al. and the results showed mean accuracy of 92.65% [24]. Agrawal and others adopted the quasi-bivariate mode decomposition (QBVMD) features with SVM classification method achieving 86.13% accuracy in binary classification, RIM-ONE r12 [25]. In the present study, iterative empirical mode decomposition (IEMD) is used in conjunction with other texture, gradient and frequency descriptors to extend these previous studies. It also has the ability to replace single-classifier approaches by a soft-voting ensemble that is able to deliver more reliable aggregation of predictive information.

III. DATASET

The RIM-ONE r12 database [26] is used in all the experiments; it is derived from fundus photographs collected at different Spanish ophthalmology centers by the Medical Image Analysis Group (MIAG). The r12 variant merges the images of release 1 with release 2, to create 505 images that comprise 255 healthy images and 250 glaucomatous images. These images are captured as 24-bit JPEG images in various spatial resolutions. Pre-partitioned folders were combined into a single data set of 505 images and stratified 10-fold cross validation was carried out. Table I outlines the distribution of the classes.

TABLE I
CLASS DISTRIBUTION IN RIM-ONE R12 DATABASE

Database	Healthy	Glaucoma	Total
RIM-ONE r12	255	250	505

To increase the training pool, each image underwent three geometric transformations: horizontal mirroring, vertical mirroring, and a 90-degree clockwise rotation. This, along with the original image, creates four variations per sample and raises the total to 2,020 images. Augmentation was applied equally across both classes before feature extraction, maintaining the original class balance.

IV. PROPOSED METHODOLOGY

Figure 1. shows the complete processing pipeline. They include six main steps: data augmentation, image preprocessing,

IEMD-based decomposition, multi-source feature extraction, dimensionality reduction, and ensemble classification.

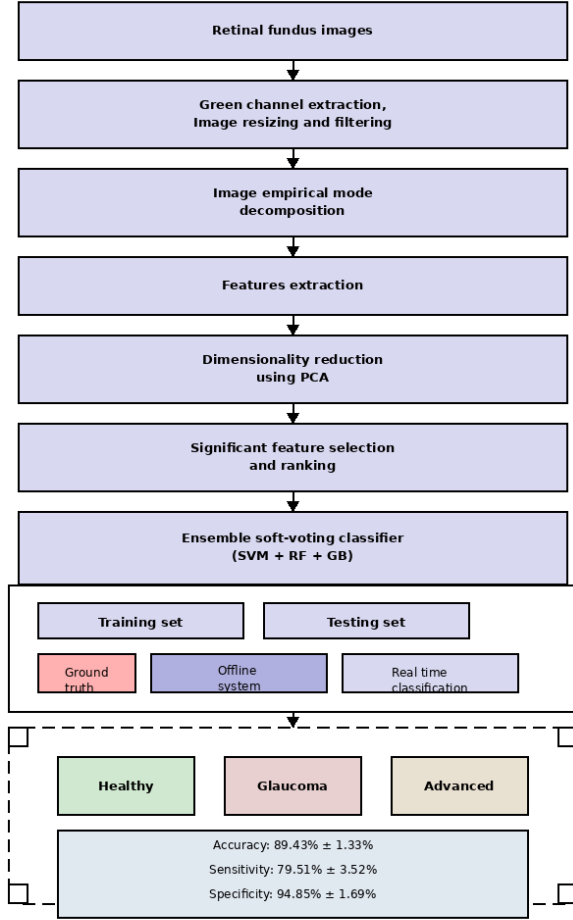


Fig. 1. Complete processing pipeline for glaucoma classification.

A. Preprocessing

All the raw fundus images are resized to 224×224 pixels with bilinear interpolation, to ensure that each image is the same size, independent of the camera used. The green channel alone is kept from the colour image since it outlines the variation between the retina and the blood vessels which give much greater detail. The photon noise is removed by applying fast non-local means denoising (where $h = 10$), and important boundaries are preserved. Contrast-limited adaptive histogram equalisation (CLAHE) with a clip limit of 4.0 and an 8×8 tile grid is then run to boost the visibility of the nerve fiber layer and the peripapillary area, especially in images that were not very clear to begin with. Lastly an unsharp masking kernel enhances the edges of the optic disc. This yields a single channel enhanced image for decomposition.

B. Image Empirical Mode Decomposition

IEMD extends empirical mode decomposition to two dimensions using the Hilbert-Huang transform, decomposing a

spatial image into a collection of intrinsic mode functions (IMFs) and a smooth residue without requiring predetermined basis functions. For a signal component to qualify as a proper IMF, two conditions must hold:

- 1) The number of extrema and zero crossings in the spatial domain must differ by at most one.
- 2) The local mean, formed from the upper and lower envelopes constructed with thin-plate smoothing splines, must equal zero everywhere.

Denoting each IMF as c_i and the final residue as r_n , the adaptive decomposition yields the reconstruction identity:

$$x(t) = \sum_{i=1}^n c_i + r_n \quad (1)$$

A Gaussian pyramid is employed in place of thin-plate spline interpolation for computational efficiency. Each pre-processed image is broken down into three IMF layers and a residue. The IMFs capture progressively lower spatial frequencies, while the residue preserves the overall illumination of the image.

C. Multi-Source Feature Extraction

The feature extraction is performed independently on five features: the three IEMD sub-bands, the image preprocessed in full size and a dedicated optic disc feature crop of size 80×80 pixels centered at approximately $(0.60W, 0.50H)$ in image coordinates. The following descriptor families are calculated for each source.

1) *GLCM Texture Features*: Three different inter-pixel distances (1, 2, 3) and four directions (0° , 45° , 90° and 135°) are used to calculate grey level co-occurrence matrix. Ten features per source are extracted: the mean and standard deviation of energy, contrast, homogeneity, correlation and dissimilarity.

2) *Entropy Measures*: Shannon entropy captures the overall randomness of grey levels. Additionally, Renyi entropy (order 2), Kapur entropy (with $\alpha = 0.5$ and $\beta = 2.0$), and Yager entropy are computed. These four measures yield four features per source.

3) *Local Binary Patterns*: The uniform LBP operator takes 3 as the radius and 24 points as the number of points to be sampled, which results in a 26 bin histogram that represents the local texture patterns that are invariant to rotations.

4) *Histogram of Oriented Gradients*: The image resolutions at each source are resized to 64×64 pixels. The HOG descriptor is calculated using 9 orientation bins, 8×8 cells and normalization performed on 2×2 blocks. Following block-average pooling, a set of 36 values summarising the gradient structure around the optic disc are extracted.

5) *Gabor Filter Responses*: Three spatial frequencies (0.1, 0.3 and 0.5 cycles per pixel) are used with four orientations, applied to both the original image and a crop around the optic disc to prevent redundant filtering in different IMF sub-bands. The mean and standard deviation of the real part of the filter response is recorded for each filter response, which results in 24 features capturing the nerve fiber textures. By combining

all the descriptors, a feature vector of length 504 is obtained per image.

D. Normalisation, Feature Selection, and Dimensionality Reduction

All features are normalised (Z score) to zero mean and unit variance. The one-way ANOVA F-statistics are computed for each feature for two binary labels and the top 100 features are kept for discarding those with low variance and class-insensitive features. Principal Component Analysis is then used to map the features onto 40 principal components which account for more than 90% of the variance.

The evaluation metrics are defined as follows:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \quad (2)$$

$$Sen = \frac{TP}{TP + FN} \times 100\% \quad (3)$$

$$Spc = \frac{TN}{TN + FP} \times 100\% \quad (4)$$

E. Ensemble Classification and Validation

Three base classifiers are trained on the 40-dimensional PCA feature space and combined via soft voting, wherein predicted class probabilities are averaged across classifiers before the final label is assigned.

1) *Support Vector Machine*: An RBF kernel SVM with class weights balanced inversely with class frequencies is employed. Optimal C and γ parameters are identified by a three-fold inner cross-validation grid search over $C \in \{10, 50, 100, 200\}$ and $\gamma \in \{\text{scale}, 0.01, 0.001\}$, executed exclusively on the training partition of each outer fold to prevent data leakage.

2) *Random Forest*: An ensemble of 300 decision trees with a maximum depth of 20, minimum samples per split of 3, and balanced class weighting is constructed by bootstrap aggregation with parallel execution.

3) *Gradient Boosting*: A gradient boosting classifier with 150 estimators, a learning rate of 0.08, a maximum tree depth of 5, and a subsample ratio of 0.8 iteratively corrects the residual prediction errors of its predecessors.

The complete ensemble is evaluated under stratified ten-fold cross-validation. Mean and standard deviation of accuracy, sensitivity, and specificity across all ten folds constitute the primary reported performance indices.

V. RESULTS AND DISCUSSION

In Table II, the fold-level and overall results are presented for the proposed ensemble with the ten-fold stratified cross-validation on the augmented RIM-ONE r12 data set. On average, the system reached 89.43% accuracy ($\pm 1.33\%$), 79.51% sensitivity ($\pm 3.52\%$), and 94.85% specificity ($\pm 1.69\%$). The high specificity means the model is not too aggressive in cases of correctly diagnosing patients as glaucoma, a good quality for screening programs, where too many false-positive cases waste valuable resources in the ophthalmic clinics. 10 Fold Cross Validation of Proposed Method is shown in Figure 2.

TABLE II
FOLD-LEVEL AND FINAL CLASSIFICATION RESULTS — RIM-ONE r12 (BINARY)

Fold	Acc (%)	Sen (%)	Spc (%)
1	89.69	79.41	95.24
2	89.18	79.41	94.44
3	89.18	76.81	96.00
4	87.11	72.46	95.20
5	91.24	79.71	97.60
6	88.66	84.06	91.20
7	91.24	85.51	94.40
8	87.69	76.81	93.60
9	89.69	81.16	94.40
10	90.72	79.71	96.80
Mean	89.43	79.51	94.85

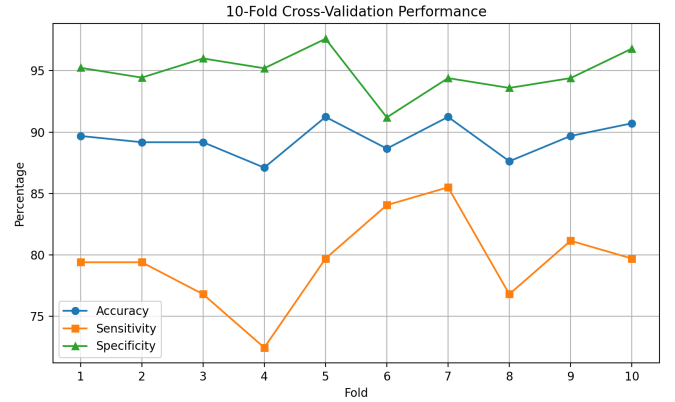


Fig. 2. Performance of the proposed ensemble model across the 10-fold cross-validation on the RIM-ONE r12 binary dataset.

The model achieved a true positive rate of 79.5% across all 10 folds and correctly classified 547 glaucomatous cases out of the 688 total. The model had a true positive rate of 79.5% across all 10 folds, and correctly classified 547 glaucomatous cases out of 688 total cases. Out of 1,252 test cases, 1,188 healthy cases were correctly identified (true negative rate of 94.9%). The difference between sensitivity and specificity indicates the challenge of detecting early-stage glaucoma, particularly when images are scaled to 224×224 . The majority of the 141 false negatives are close calls involving cases of structural cupping where the IEMD sub-band or texture features have only slightly changed. Confusion Matrix is represented in Figure 3.

The proposed method is compared with the various existing methods evaluated in the RIM-ONE r12 database, as shown in Table III. The proposed system is much better than Maheshwari et al.'s EWT method with an accuracy of 8.77 percentage points, outperforming DWT-EWT hybrid by Kirar et al. by 5.83 points and outperforming QBVMD pipeline of Agrawal et al. by 3.30 points. The specificity is slightly higher (94.85% vs. 87.84%) and marginally lower for accuracy (89.43% vs. 90.76%), which is a desirable balance when screening the population as there is a need to minimise the number of

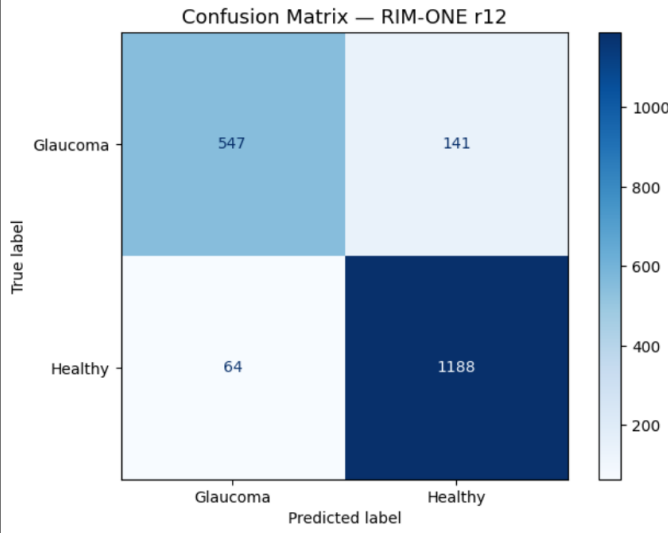


Fig. 3. Aggregated confusion matrix across ten folds of stratified cross-validation on the RIM-ONE r12 dataset.

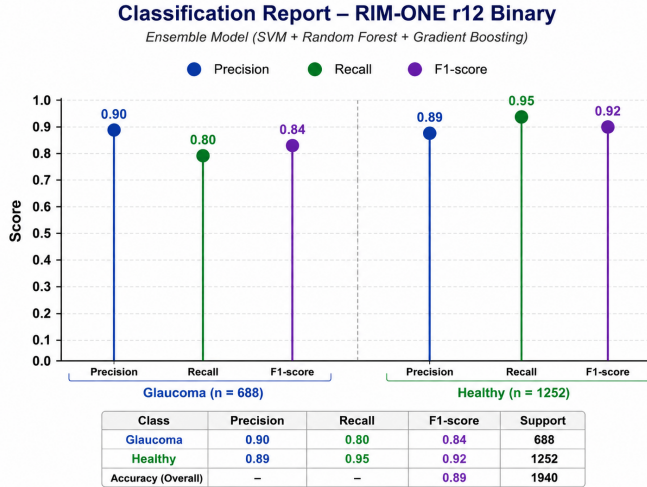


Fig. 4. Lollipop chart illustrating the classification performance of the proposed ensemble model on the RIM-ONE r12 binary dataset.

false positives. The precision we report for IEMD in [18] cannot be directly compared because a private set of 232 additional images was used and a simple LS-SVM with only 14 PCA components was applied; in this work, only the public benchmark is considered and the results of the classification performance of the glaucoma and healthy sample images are shown in Figure 4, Comparison of state of art methods is represented in Figure 5.

The fold-wise accuracy is fairly stable with little variance ($SD = 1.33\%$) demonstrating the low standard deviation, indicating stable performance across different train-test splits. The optimal hyperparameters for the SVM, C and γ , were always chosen as $C = 100$ and $\gamma = \text{scale}$, respectively, by inner cross-validation. When the optic disc area was targeted and GLCM and HOG features were extracted from the area around the

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS ON RIM-ONE r12

Method	Acc (%)	Sen (%)	Spc (%)
Maheshwari et al. [14]	80.66	78.00	88.23
Kirar et al. [21]	83.60	86.40	80.80
Agrawal et al. [25]	86.13	84.80	87.43
Yi et al. [16]	86.80	86.55	93.84
Proposed (Ensemble)	89.43	79.51	94.85

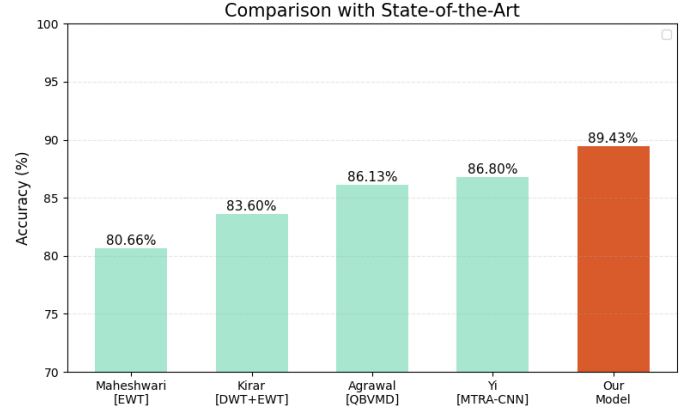


Fig. 5. Comparison between proposed method's accuracy and former methods using RIM-ONE r12 benchmark.

optic disc, clear improvement in performance was achieved, possibly because these features provide information about the cup-to-disc ratio which is important in the diagnosis of glaucoma. While a large proportion of the existing sensitivity and specificity difference may be explained by the challenge of detecting early glaucomatous changes from $2D\ 224 \times 224$ images in the absence of 3D structural depth information, the underlying reasons for this remain unknown.

VI. CONCLUSION

An automated system was built to classify retinal fundus images as either glaucomatous or healthy, using the RIM-ONE r12 dataset as a benchmark. The approach combines IEMD for multi-scale image decomposition with a broad set of feature extraction methods: GLCM texture descriptors, LBP, HOG, Gabor filter responses, and four entropy measures. These features are derived from both the IEMD sub-bands and targeted regions including the full preprocessed image and an optic disc crop. ANOVA-based feature selection and PCA dimensionality reduction retain only the most discriminative components. A soft-voting ensemble of SVM, Random Forest, and Gradient Boosting models, validated with ten-fold stratified cross-validation, achieves 89.43% accuracy, 79.51% sensitivity, and 94.85% specificity, outperforming several existing approaches on the same dataset. The high specificity is especially valuable for large-scale population screening. Future work will extend the framework to three-stage glaucoma grading, incorporate deep convolutional features via transfer learning, and validate

the system prospectively across multiple clinical centers with larger and more diverse image sets.

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